Jio Platforms Limited

VAULT SERVER SETUP GUIDE

Consent Management System

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Version History

Date	Contributor	Contribution	Version	State
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Table of Contents

Version History *1*

1. Vault Server Setup Guide *4*

2. Installing Vault (Linux) *4*

2.1 Download & Install Vault **4**

2.2 Create Vault System User **4**

3. Directory Structure (with Commands) *4*

3.1. Create Required Directories **4**

4. Generate TLS Certificate (Optional but Recommended) *5*

*5. Installation Verification*** *5*

6. Vault File Structure *5*

7. Vault Configuration (HCL) *5*

8. Starting Vault Server *6*

9. Environment Variable Setup *6*

10. Initialize Vault *6*

11. Unseal Vault *7*

12. Login to Vault *7*

13. Enable KV Secret Engine *7*

14. Create Policy (Example) *7*

15. Create Token with Policy *7*

16. Create Systemd Service (Recommended) *8*

17. Accessing Vault UI *8*

18. Next Steps (Optional Enhancements) *8*

19. Command Reference (Step-by-Step) *9*

1. Install Vault 9
2. Create Vault System User 9
3. Create Directory Structure 9
4. Set Ownership 9
5. Generate TLS Certificates 9
6. Start Vault Server Manually 9
7. Export VAULT_ADDR 9
8. Check Status 9
9. Initialize Vault 9
10. Unseal Vault (Run 3 Times) 10
11. Login 10

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12. Enable KV Secrets Engine 10
13. Add a Test Secret 10
14. Create Policy File 10
15. Apply Policy 10
16. Create Token for Policy 10
17. Create Systemd Service 10
18. Enable & Start Service 10

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1. Vault Server Setup Guide

This document provides a clear, step-by-step guide for installing, configuring, initializing, and running HashiCorp Vault on a Linux server. It is based on your working configuration and commands executed in your environment.

2. Installing Vault (Linux)

2.1 Download & Install Vault

- `wget https://releases.hashicorp.com/vault/1.19.4/vault_1.19.4_linux_amd64.zip`
- `unzip vault_1.19.4_linux_amd64.zip`
- `sudo mv vault /usr/local/bin/`
- `vault -version`

2.2 Create Vault System User

- `sudo useradd --system --home /etc/vault --shell /bin/false vault`

3. Directory Structure (with Commands)

3.1. Create Required Directories

-
```
sudo mkdir -p /etc/vault/config
sudo mkdir -p /etc/vault/tls
sudo mkdir -p /etc/vault/policies
sudo mkdir -p /etc/vault/data
```

Recommended structure:

```
/etc/vault/
├── config/
│   ├── config.hcl
│   ├── listeners.hcl
│   ├── storage.hcl
│   └── telemetry.hcl
├── tls/
│   ├── vault.crt
│   └── vault.key
└── policies/
    ├── admin-policy.hcl
    └── app-policy.hcl

/etc/vault/data/
```

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4. Generate TLS Certificate (Optional but Recommended)

- `openssl genrsa -out /etc/vault/vault.key 2048`
- `openssl req -new -key /etc/vault/vault.key -out /etc/vault/vault.csr`
- `openssl x509 -req -days 365 -in /etc/vault/vault.csr -signkey /etc/vault/vault.key -out /etc/vault/vault.crt`

5. Installation Verification**

- Ensure Vault binary is present:

  `/usr/local/bin/vault -version`

- If Vault is not in PATH, run it using:

  `./vault -version`

6. Vault File Structure

- Vault Binary:

  `/usr/local/bin/vault`

- Config Directory:

  `/etc/vault/config/`

- Data Directory:

  `/etc/vault/data`

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7. Vault Configuration (HCL)

Your working config file:

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```
storage "file" {
  path = "/etc/vault/data"
}

listener "tcp" {
  address       = "0.0.0.0:8200"
  tls_disable   = "true"
}

api_addr = "http://<SERVER-IP>:8200"
cluster_addr = "http://<SERVER-IP>:8201"
ui = true
```

Replace <SERVER-IP> with your server IP (e.g., 10.173.184.27).

8. Starting Vault Server

- Start Vault manually:

  ```
  vault server -config=/etc/vault/config/
  ```

- If mlock error occurs, add this to config:

  ```
  disable_mlock = true
  ```

9. Environment Variable Setup

In a new terminal:

```
export VAULT_ADDR="http://<SERVER-IP>:8200"
```

Check status:

```
vault status
```

10. Initialize Vault

- Vault must be initialized once:

  ```
  vault operator init
  ```

- This produces: - 5 Unseal Keys - Root Token

  Store all keys safely.

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11. Unseal Vault

- Run the below for 3 keys:

```
vault operator unseal <unseal-key-1>
vault operator unseal <unseal-key-2>
vault operator unseal <unseal-key-3>
```

- Check:

```
vault status
```

- Should show:

```
Initialized: true
Sealed: false
```

12. Login to Vault

Use the root token:

```
vault login <root-token>
```

13. Enable KV Secret Engine

```
vault secrets enable -path=secret kv-v2
```

Test:

```
vault kv put secret/test key1=value1
vault kv get secret/test
```

14. Create Policy (Example)

- Create a read-only policy:

```
path "secret/data/*" {
  capabilities = ["read"]
}
```

- Apply:

```
vault policy write readonly readonly.hcl
```

15. Create Token with Policy

```
vault token create -policy=readonly
```

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16. Create Systemd Service (Recommended)

- Create file:

  ```
  /etc/systemd/system/vault.service
  ```

- Content:

  ```
  [Unit]
  Description=HashiCorp Vault
  Requires=network-online.target
  After=network-online.target

  [Service]
  User=root
  Group=root
  ExecStart=/usr/local/bin/vault server -config=/etc/vault/config/
  ExecReload=/bin/kill --signal HUP $MAINPID
  KillSignal=SIGTERM
  Restart=on-failure
  LimitNOFILE=65536

  [Install]
  WantedBy=multi-user.target
  ```

- Enable & start:

  ```
  systemctl daemon-reload
  systemctl enable vault
  systemctl start vault
  systemctl status vault
  ```

17. Accessing Vault UI

- Open browser:

  ```
  http://<SERVER-IP>:8200
  ```

- Login using root token.

18. Next Steps (Optional Enhancements)

- Enable TLS certificates
- Configure AppRole auth for applications
- Setup Kubernetes auth
- Use Vault Agent for auto-auth + templates
- Create namespaces for multi-team usage

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19. Command Reference (Step-by-Step)

1. Install Vault

```
wget https://releases.hashicorp.com/vault/1.19.4/vault_1.19.4_linux_amd64.
zip
unzip vault_1.19.4_linux_amd64.zip
sudo mv vault /usr/local/bin/
vault --version
```

2. Create Vault System User

```
sudo useradd --system --home /etc/vault --shell /bin/false vault
```

3. Create Directory Structure

```
sudo mkdir -p /etc/vault/config
sudo mkdir -p /etc/vault/tls
sudo mkdir -p /etc/vault/policies
sudo mkdir -p /opt/idam/vault/data
```

4. Set Ownership

```
sudo chown -R vault:vault /etc/vault
sudo chown -R vault:vault /opt/idam/vault
```

5. Generate TLS Certificates

```
openssl genrsa -out /etc/vault/tls/vault.key 2048
openssl req -new -key /etc/vault/tls/vault.key -out /etc/vault/tls/vault.c
sr
openssl x509 -req -days 365 -in /etc/vault/tls/vault.csr -signkey /etc/vau
lt/tls/vault.key -out /etc/vault/tls/vault.crt
```

6. Start Vault Server Manually

```
vault server -config=/etc/vault/config/
```

7. Export VAULT_ADDR

```
export VAULT_ADDR="http://<SERVER-IP>:8200"
```

8. Check Status

```
vault status
```

9. Initialize Vault

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```
vault operator init
```

10. Unseal Vault (Run 3 Times)

```
vault operator unseal <UNSEAL-KEY>
```

11. Login

```
vault login <ROOT-TOKEN>
```

12. Enable KV Secrets Engine

```
vault secrets enable -path=secret kv-v2
```

13. Add a Test Secret

```
vault kv put secret/test key=value
vault kv get secret/test
```

14. Create Policy File

```
cat <<EOF > readonly.hcl
path "secret/data/*" {
  capabilities = ["read"]
}
EOF
```

15. Apply Policy

```
vault policy write readonly readonly.hcl
```

16. Create Token for Policy

```
vault token create -policy=readonly
```

17. Create Systemd Service

```
sudo tee /etc/systemd/system/vault.service > /dev/null <<EOF
[Unit]
Description=HashiCorp Vault
Requires=network-online.target
After=network-online.target

[Service]
User=vault
Group=vault
ExecStart=/usr/local/bin/vault server -config=/etc/vault/config/
ExecReload=/bin/kill --signal HUP $MAINPID
KillSignal=SIGTERM
Restart=on-failure
LimitNOFILE=65536

[Install]
WantedBy=multi-user.target
EOF
```

18. Enable & Start Service

```
systemctl daemon-reload
systemctl enable vault
```

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```
systemctl start vault
systemctl status vault
```

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